Requirements Table

Requirements description	Type
Capture customer name, delivery address and phone number	Functional
Validate the number of pizzas ordered between 1 and 6	Functional
Track the size of each pizza by size (small, medium and large)	Functional
Track the number of extra toppings and the cost (£0.75 for 1, £1.35 for 2, £2.00 for 3, £2.50 for 4+)	Functional
Check if delivery is required and add the flat £2.50 charge if needed	Functional
Calculate if the order is over £20 before the delivery fee is added and apply the 10% discount	Functional
Output a clear and personalised itemised receipt breaking down the order and the final total.	Functional

Variables

	Variable	Input/Output/Process
cName	Customer name	Input
cAddress	Address	Input
cPhone	Phone number	Input
NumPizza	Number of Pizzas	(1<=x<=6) input
SizePizza	Size of Pizzas	(s,m,l) input
NumToppings	Number of toppings	(0<=x<=4) input
Discount	Discount	(True False) input
Deliveries	Delivery required	(True False) input
CusDetail	Customer Detail	Input
CostPizza	Cost of each pizza	Process
SubtotalPre	Order subtotal before discount and delivery	Output
DiscApply	Any discount applied	Process
DelivApply	Any delivery charge	Process
FinalTotal	Final total cost of the order	Output

Functions

Functions	Varibles used	Formula/storage types

cDetails	cName, cAddress, cPhone	Store in array
nPizza	numPizza	Store the number of pizzas between 1-6
sPizza	NumPizza,sizePizza, costPizza	Check numPizza and assign sizePizza in a loop
pricePizza	CostPizza, sizePizza	Check sizePizza and assign costPizza

Def pizza*fn(pizza*-order): (list) -> (list, R)

_ _ init _ _(pizza) = pizza

Format: {Pizza-*1,* Pizza-*2, ..., Pizza*-6} -where Pizza-X can be Empty SET {}

or {size, extra-toping-1, extra-topping 2,...}